

Toward Expert-Level Medical Question Answering (Med-PaLM 2)

Singhal, K., Tu, T., Gottweis, J., et al. (2025). "Toward expert-level medical question answering with large language models." *Nature Medicine* 31. Published online 8 January 2025. Open access (CC BY).

ASSIGNED READING

Read the abstract, the prompting-strategy methods, and the human-evaluation results ~45 min
If short on time, read this summary in full and skim the paper —
New to the vocabulary? Read the appendix at the end of this brief first ~10 min
» ~45 min • this (detailed) summary + appendix ~25 min

Which version to read. Read the *Nature Medicine* paper linked below — it is the peer-reviewed version of record, and it is open access, so there is no paywall. An earlier preprint (arXiv 2305.09617, May 2023) is also circulating; it is largely the same work, but it omits two things we use in class: the chain-of-retrieval grounding pipeline, and the real-world bedside-consultation pilot study. If you have the preprint, use the *Nature* version instead.

Note on currency. Med-PaLM 2 has since been overtaken by newer medical models, and its base model is not the latest. That is not the point. Read it as the worked example for Capsule 3: a real, high-stakes system where careful prompting and rigorous evaluation — not a bigger model — closed the gap to expert answers. This is a detailed summary because the methods matter more than the leaderboard number.

KEY TAKEAWAY

A general model can be pushed to near-expert medical question answering mostly through how you prompt it and how carefully you evaluate it. Med-PaLM 2 reached about 86.5% on MedQA (US Medical Licensing Examination-style questions) — a gain of over 19 points on the previous version — by combining an improved base model with medical fine-tuning and, critically, inference-time prompting strategies including a new one the team calls ensemble refinement. Just as important is how they proved it: physicians compared long-form answers on nine clinical axes, and preferred Med-PaLM 2's answers to other physicians' answers on eight of those nine. Yet in a real bedside-consultation pilot, specialist answers were still preferred overall — the honest result that keeps this from being a victory lap.

WHY IT MATTERS FOR THIS COURSE

This is Leg A of the Class-3 case — the half that proves the capability. Capsule 2 taught you a ladder of prompting techniques and the discipline of measuring a prompt against an eval set you wrote. This paper is that ladder, climbed to the top, in a domain where a confident wrong answer can kill someone: few-shot examples, chain of thought, self-consistency, then ensemble refinement stacked on top; and a nine-axis physician rubric instead of a single accuracy score. Then comes the turn. Every one of those techniques is paid for in tokens, on every question, forever — which is exactly the question Leg B (the AI Wars case, briefed live in class) puts to a business. Read this one asking: what did that accuracy cost?

Feeds Class 3: C2 (the prompting ladder, at full stretch) • C3 Leg A — capability, then the hinge into AI Wars economics.

MAIN POINTS

The problem

- **Medical answers are long-form, not multiple choice.** Benchmarks like MedQA test whether a model can pick the right option. Real clinical and consumer questions need a paragraph that is accurate, complete, unbiased and safe — and no single percentage captures that.
- **The starting point was a general model.** Med-PaLM 2 builds on an improved base model (PaLM 2) plus medical domain fine-tuning — but the headline gains come from inference-time prompting strategy, not from training a medical model from scratch.

How they improved it — the prompting stack

- **Few-shot and chain of thought (CoT).** Worked medical examples in the prompt, plus step-by-step reasoning before the answer — the same rungs you used in Capsule 2.
- **Self-consistency (SC).** Sample many reasoning paths at a non-zero temperature and take the majority answer. Trades compute for reliability on hard questions.
- **Ensemble refinement (ER) — the new move.** A two-stage generalisation of self-consistency. First the model produces several candidate explanations-and-answers by temperature sampling. Then it is conditioned on the original prompt, the question, and all of those generations concatenated, and asked to produce one refined explanation and answer. So it does not merely vote — it reads its own attempts and rewrites.

- **Chain of retrieval — grounding by search.** For difficult bedside-consultation questions, a step-by-step pipeline: generate an initial answer, break it into individual claims, generate search queries for those claims, retrieve external medical sources, then verify and revise. This is retrieval-augmented generation (RAG) — the Week-2 pipeline — reappearing as a safety mechanism.
- **The result.** About 86.5% on MedQA, over 19 points above the original Med-PaLM, with state-of-the-art or near-state-of-the-art results across MedMCQA, PubMedQA and MMLU clinical topics.

How they evaluated it — the part to copy

- **Nine axes, not one number.** Physicians rated long-form answers on four “high-quality answer traits” — reflects consensus, reading comprehension, knowledge recall, reasoning — and five “potential answer risks” — inaccurate or irrelevant information, omitted information, evidence of demographic bias, extent of possible harm, and likelihood of harm.
- **Head-to-head against physicians.** In a pairwise ranking of 1,066 consumer medical questions, physician raters preferred Med-PaLM 2’s answers to other physicians’ answers on eight of the nine axes.
- **Adversarial sets, on purpose.** Two adversarial question datasets — one general, one health-equity focused — were introduced specifically to probe where the model breaks, rather than to flatter it. See the dedicated section below; this is the part most teams skip.
- **Raters were plural and international.** Fifteen physician raters — six based in the United States, four in the United Kingdom, five in India — spanning family medicine and general practice, internal medicine, cardiology, respiratory, paediatrics and surgery. Six laypeople (none with a medical background) rated helpfulness and relevance separately. Raters were blinded to which answer came from the model and rated independently, without conferring.

Adversarial testing — the part most teams skip

- **What “adversarial” means here.** Not hacking or jailbreaking. These are ordinary-looking medical questions deliberately curated to elicit answers with potential for harm or bias — the questions where a fluent, confident, wrong answer does the most damage. The team wrote them before measuring, so the evaluation could not quietly avoid its own weak spots.
- **Two datasets, 240 questions.** Adversarial (General) — 58 questions covering health disparities, drug use, alcohol, mental health, COVID-19, obesity, suicide and medical misinformation. Adversarial (Health Equity) — 182 questions built around healthcare access (insurance, reaching a hospital or a primary-care provider), care quality (patient experience, coordination) and social and environmental factors (working and living conditions, food access, transport). The second set was released publicly as the Open-ended Medical Adversarial Queries (OMAQ) dataset.
- **Designed to be badly asked.** The health-equity set deliberately mixes implicit and explicit adversarial queries. Many are implicit requests for medical advice rather than well-formed medical questions — because that is how real people actually type. A benchmark made only of clean, well-formed questions flatters the model by testing it in conditions it will never meet.
- **Deliberately named prior blind spots.** One example the paper calls out: racial bias baked into clinical calculators for kidney function. The adversarial set was curated from the health-equity-in-machine-learning literature, so it tests for failure modes the field already knows about rather than waiting to discover them in production.
- **Rated harder than the main set.** Answers to adversarial questions were quadruple-rated — four independent physician raters each — against triple rating for the main MultiMedQA sample. Where you expect disagreement, buy more raters.
- **The result — and the tell.** On the adversarial sets Med-PaLM 2 was rated significantly higher quality than Med-PaLM on every axis ($P < 0.001$), in both the general and the health-equity subsets, often by substantial margins. But notice the tell: on adversarial questions the median Med-PaLM 2 answer ran to 964 characters, against 794 on the ordinary consumer set — and 337.5 characters for a physician’s answer. When the question got harder, the model got longer. That is the same padding instinct that cost it the ninth axis.
- **The authors’ own limitation.** They state it plainly: the adversarial evaluation is “relatively limited in scope and should not be interpreted as a comprehensive assessment of safety, bias and equity considerations.” 240 questions is a probe, not a safety case. Note what they did with that — they published the limitation rather than the headline.
- **What to copy.** Your Capsule 2 eval harness had three adversarial cases out of ten for exactly this reason. Write the cases that are designed to break your system before you need them, rate the contentious ones more than once, and report what you did not cover.

The honest result

- **A real-world pilot, not just benchmarks.** On bedside-consultation questions that arose during actual care delivery, specialists preferred Med-PaLM 2's answers to generalist physicians' answers about 65% of the time, and generalists rated them roughly equally.
- **But specialists still won overall.** Specialist-written answers were still preferred overall — while both specialists and generalists rated Med-PaLM 2's answers as about as safe as physician answers. Strong, useful, and not yet a replacement.
- **Why that matters for us.** This is what a mature claim looks like: a real number, a named limitation, and no suggestion that the benchmark settles the clinical question.

READING FOCUS QUESTIONS — bring your answers to class

There are no wrong answers here — these surface how you think, not what you recall.

- 1 • The big gains came from prompting strategy, not a new model. What does that suggest about where to spend your effort when your first prompt underperforms?
Clue. Think about what each option actually costs you. A better base model is someone else's release schedule; a better prompt is an afternoon. Then think about the ceiling: prompting cannot invent knowledge the model does not have. Where is the line between "my prompt is under-specified" and "this model cannot do this task"? Capsule 2's ladder — be clear, few-shot, chain of thought, structured output, then sampling — is a spending order, cheapest rung first.
- 2 • Ensemble refinement asks the model to read several of its own attempts before answering once. Roughly how many times more expensive is that than a single answer — and in which situations would you actually pay it?
Clue. Do the arithmetic from the appendix: the paper used 11 samples in the first stage and 33 in the second, and every one of those second-stage calls carries all eleven earlier answers in its input. So think in two directions — number of calls, and length of each call. A ballpark of "tens of times" a single answer is the right order of magnitude. Then ask the business question: what is the cost of one wrong answer in your setting? Pay for the technique where a mistake costs more than the tokens; do not pay for it on a chatbot that recommends restaurants.
- 3 • The team refused to judge quality on accuracy alone and built a nine-axis rubric. For an AI answer in your industry, which two or three axes beyond "correct" would you insist on measuring?
Clue. Start from how your organisation gets hurt. Some candidates that transfer well: completeness (what was omitted); traceability (can I see the source); tone or appropriateness for the reader; regulatory defensibility (would this survive an audit); bias across customer segments; and cost or latency per answer. Pick axes you could actually score — "trustworthy" is not measurable, "cites a source that exists" is. And notice which axis Med-PaLM 2 lost on, then ask whether your industry would forgive that failure or be sunk by it.
- 4 • Specialists still preferred specialist answers, yet rated the model as safe. If you ran a hospital, what would you do with a system like that — deploy it, restrict it, or wait?
Clue. Resist the binary. The realistic answers live in between: deploy it behind a clinician rather than in front of a patient; restrict it to a narrow question type where you have measured it; use it to draft and require a human to sign. Then ask what would move you — which number, at what level, would change your decision? That is the useful managerial output of this case, and it is the same question the AI Wars case will ask about money instead of safety.

APPENDIX — THE TECHNIQUES, EXPLAINED

Seven terms do most of the work in this paper. Several will be new to you, and the paper assumes you already know them. Each entry below gives the plain-language idea first, then the technical statement — and, where the paper reports it, the exact setting Med-PaLM 2 used. Read this section before the paper, not after.

One distinction to hold on to. Six of these seven are things you do at inference time — at the moment you ask the question. They change the prompt, not the model, so you can try them this afternoon and they cost tokens on every single request, forever. Only fine-tuning changes the model itself: expensive once, then cheap per request. That difference is the entire bridge from this case into the economics case.

1 • Zero-shot, one-shot and few-shot prompting

- **Plain.** How many worked examples you put in the prompt before your actual question. Zero-shot means you just ask. One-shot means you show one solved example first. Few-shot means you show several. You are not teaching the model anything permanent — you are showing it the shape of the answer you want, in the same breath as the question. (You will hear "single-shot" used loosely for either zero-shot or one-shot; the paper uses "zero-shot" and "few-shot", so use those.)
Technically. Formally, in-context learning: example input-output pairs are prepended to the final input, and the model conditions its next-token predictions on them. Nothing is written back to the weights — close the session

and the effect is gone. Cost: every example is re-sent, and re-billed, on every request. Med-PaLM 2 used the same few-shot prompts as the original Med-PaLM, and used a zero-shot prompt for the first step of chain of retrieval.

2 • Chain of thought (CoT)

- **Plain.** Asking the model to work through the reasoning step by step before it commits to an answer, instead of blurting the answer out. Medical questions usually need several linked steps, and a model that writes those steps down gets more of them right.

Technically. Each few-shot example is augmented with a step-by-step explanation leading to the final answer. Because the model is autoregressive — every token it generates becomes part of the input for the next token — the written-out reasoning becomes context the model can condition on. The intermediate steps are not merely a display for you; they are working memory. Cost: the output gets several times longer, and output tokens are the expensive kind. The paper hand-crafted its CoT prompts for medical questions.

3 • Temperature

- **Plain.** A dial from about 0 to 1 (sometimes higher) controlling how adventurous the model is when picking the next word. At 0 it always takes the most likely word — same question, same answer, every time. Turn it up and it sometimes picks the second- or third-most-likely word, so you get variety, and sometimes a better idea, and sometimes nonsense.

Technically. The model outputs a score (logit) for every possible next token; those scores are divided by the temperature T before the softmax that turns them into probabilities. Low T sharpens the distribution towards the top token — at $T \rightarrow 0$ it becomes greedy decoding, deterministic. High T flattens it, raising the chance of low-probability tokens. Temperature is the engine underneath both techniques that follow: without it, sampling the model repeatedly would just return the identical answer. Med-PaLM 2's long-form consumer answers were sampled at temperature 0.0 — deterministic — while the multiple-choice sampling strategies used a non-zero temperature by design.

4 • Self-consistency (SC)

- **Plain.** Ask the same question many times with the temperature turned up, then take the answer that came back most often. A majority vote by one model against itself. It works because there is usually more than one valid route to a right answer, but many different routes to a wrong one.

Technically. Sample n independent chain-of-thought reasoning paths, extract the final answer from each, and take the majority (or plurality) vote — marginalising over reasoning paths rather than trusting a single sample. Med-PaLM 2 used 11 samplings with CoT prompting. Cost: 11 samples means roughly $11\times$ the tokens of one answer. Limitation: it only works when answers can be tallied — multiple choice, a label, a number. You cannot majority-vote over paragraphs.

5 • Ensemble refinement (ER) — the new move in this paper

- **Plain.** Two stages. First, generate several candidate answers-with-reasoning by asking repeatedly at a non-zero temperature. Then hand all of them back to the model in one prompt and say: here are your own attempts — read them and write one better answer. It does not merely count votes, it reads the strengths and weaknesses of its own drafts and rewrites. Think of it as a writer reviewing several drafts rather than a committee taking a show of hands.

Technically. Stage 1: given a few-shot CoT prompt and the question, produce multiple generations stochastically via temperature sampling. Stage 2: condition the model on the original prompt, the question, and the concatenated stage-1 generations, and prompt it for a refined explanation and answer. The paper runs stage 2 multiple times and takes a plurality vote over those refined answers. It is a generalisation of self-consistency — aggregation by a language model instead of by counting — and it builds on CoT and self-refine. Med-PaLM 2 used 11 samplings in stage 1 and 33 in stage 2. Note the input cost: every one of those 33 calls carries all 11 stage-1 answers in its prompt. Unlike SC, ER can in principle aggregate long-form answers, not just multiple-choice ones — but the paper applied it only to multiple choice, and says why: “the resource cost of approaches requiring repeated samplings.”

6 • Chain of retrieval

- **Plain.** Do not trust the model's memory — make it go and check. Draft an answer, break the draft into individual factual claims, turn each claim into a search query, search the web, summarise what comes back, then write the final answer from those summaries. The Week-2 retrieval pipeline, reappearing here as a safety mechanism rather than a convenience.

Technically. A six-step pipeline: (1) generate an initial answer with a zero-shot prompt; (2) separate it into individual claims for verification; (3) generate search queries for those claims; (4) retrieve studies and websites via Google search; (5) summarise each document; (6) generate the final answer from the question plus the concatenated summaries. Steps 1, 2, 3 and 6 are separate model inferences; step 5 is one inference per retrieved document. One design detail worth stealing: the initial answer from step 1 is deliberately excluded from the step-6 prompt, to stop the model anchoring on its own ungrounded draft. The authors distinguish this from standard retrieval-augmented generation, which searches a fixed corpus and embedding space; here the retrieval is driven

by claims the model itself extracted. Cost: one question becomes many calls, and grows with the number of documents retrieved.

7 • Fine-tuning (here, instruction fine-tuning)

- **Plain.** Actually changing the model, by continuing to train it on examples from your domain. Everything above changes what you say to the model; this changes the model. It costs a lot once, then costs nothing extra per question — the opposite trade to every technique above.

Technically. Instruction fine-tuning updates the base model's weights on curated instruction-response data. Med-PaLM 2 was fine-tuned on the training splits of MultiMedQA — MedQA, MedMCQA, HealthSearchQA, LiveQA and MedicationQA — as a single “unified” model, with dataset mixture ratios tuned empirically: they started proportional to each dataset's size, then overweighted the datasets with more high-quality, diverse examples. Data mixture is a design decision, not a detail. A multiple-choice-only variant scored better on multiple-choice benchmarks — a reminder that fine-tuning optimises for whatever you fine-tune on. Cost profile: high fixed cost, near-zero marginal cost, and it locks you to a model version.

The cost ladder — carry this into Capsule 3

Ranked by what they add to a single answer: a plain zero-shot question is 1×. Few-shot adds input tokens on every call. Chain of thought multiplies output tokens. Self-consistency multiplies the whole call — 11× here. Ensemble refinement multiplies it again, and inflates each call's input — tens of times a single answer. Chain of retrieval turns one question into a variable number of calls plus search. Fine-tuning is the only one that is paid up front instead of per question. Every technique that made Med-PaLM 2 better made each answer more expensive, and that bill arrives on every request for as long as the system runs. That is precisely the question the second case puts to a business.

SOURCE LINKS

Paper (Nature Medicine, open access) — read this one · [nature.com/articles/s41591-024-03423-7](https://www.nature.com/articles/s41591-024-03423-7)

Earlier preprint (superseded) · arxiv.org/abs/2305.09617

Project background · sites.research.google/med-palm

CAVEATS

1. This is a detailed summary of a research paper, provided so you can grasp the method without reading all of it. It is a map, not the territory — the paper is open access, so read the methods and evaluation sections for yourself; they are the authority.
2. This summary was generated and fact-checked using Anthropic models against the Nature Medicine paper. It paraphrases; it does not reproduce the article. Flag anything that looks wrong.